Assignment - 1

Underground Space Technology (MND407)

Max. Marks: 10

Instructions: Last date of submission of Assignment - ***<u>18th February to 20 February 2026</u>***

Q. No.	Question	Marks
1	During an excavation for a tunnel, 250 m below ground level, a highly fractured siltstone rock mass with two major joint sets and many random fractures is encountered. The RQD from the rock cores is 40% and the average Uniaxial Compressive Strength (UCS) is 70 MPa. The joints with average spacing of 75 mm are rather continuous with high persistence with apertures of 3 - 5 mm, and they are filled with silty clay. The joint surface walls are highly weathered, undulating, and slickensided. There is some water inflow into the tunnel estimated as 15 L/min per 10 m of tunnel length, with some outwash of joint fillings. Assume a unit weight of 28 kN/m^3 for the rock mass. (a) Estimate RMR and *Q* values. (b) Check the empirical correlations relating RMR and Q, in light of the estimated RMR and Q values. (c) Estimate the Equivalent Rock Mass Modulus (E_m) using the calculated RMR and Q values. (d) Assess the stability of a 10 m wide proposed excavation in the above given rock conditions. (e) Draw a clear conceptual failure mechanism diagram for the proposed 10 m wide tunnel.	10